



DATABASE SYSTEM LABS

- ADVANCED SQL: PROCEDURE

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OBJECTIVES

- Stored Procedure
- Tasks

drop procedure if exists precourse_cnt_proc;

DELIMITER \$\$

create procedure precourse_cnt_proc(
in acourseid varchar(8), out acnt integer)

deterministic

reads sql data

begin

select count(*) into acnt from prereq

where course_id=acourseid;

end\$\$

DELIMITER ;

CALL PROCEDURE

- In mysql workbench

```
set @course_id='958';
```

```
call precourse_cnt_proc(@course_id,@cnt);
```

```
select @cnt as cnt;
```

CALL PROCEDURE

- In sql functions and procedures

```
declare course_id varchar(10);
```

```
declare cnt integer;
```

```
select '958' into course_id;
```

```
call precourse_cnt_proc(course_id, cnt);
```

CALL PROCEDURE IN PYTHON3

```
import pymysql

conn = pymysql.connect(host='localhost',
user='mydbuser',password='iammydbuser',database='dblabtest',charset='utf8')

cursor = conn.cursor()

aInCourse_id = '958'

aCnt = -1

cursor.callproc('precourse_cnt_proc',(aInCourse_id,aCnt))

cursor.execute('select @_precourse_cnt_proc_0 as Course_ID,@_precourse_cnt_proc_1 as Cnt')

retdata = cursor.fetchone()

if retdata:

    aCnt = retdata[1]

conn.commit()

cursor.close()

conn.close()

print('The result:%d' %(aCnt))
```

drop procedure if exists precourse_cnt_proc2;

DELIMITER \$\$

create procedure precourse_cnt_proc2(

in acourseid varchar(8), out acnt integer)

deterministic

reads sql data

begin

select count(*) into acnt from prereq where course_id=acourseid;

select * from prereq where course_id=acourseid;

end\$\$

DELIMITER ;

CALL PROCEDURE IN PYTHON3

```
import pymysql

conn = pymysql.connect(host='localhost',
user='mydbuser',password='iammydbuser',database='dblabtest',charset='utf8')

cursor = conn.cursor()

aInCourse_id = '958'

aCnt = -1

cursor.callproc('precourse_cnt_proc2',(aInCourse_id,aCnt))

retdata1 = cursor.fetchall()

print(retdata1)

cursor.execute('select @_precourse_cnt_proc2_0 as Course_ID,@_precourse_cnt_proc2_1 as Cnt')

retdata2 = cursor.fetchone()

if retdata2:

    aCnt = retdata2[1]

cursor.close()

conn.close()

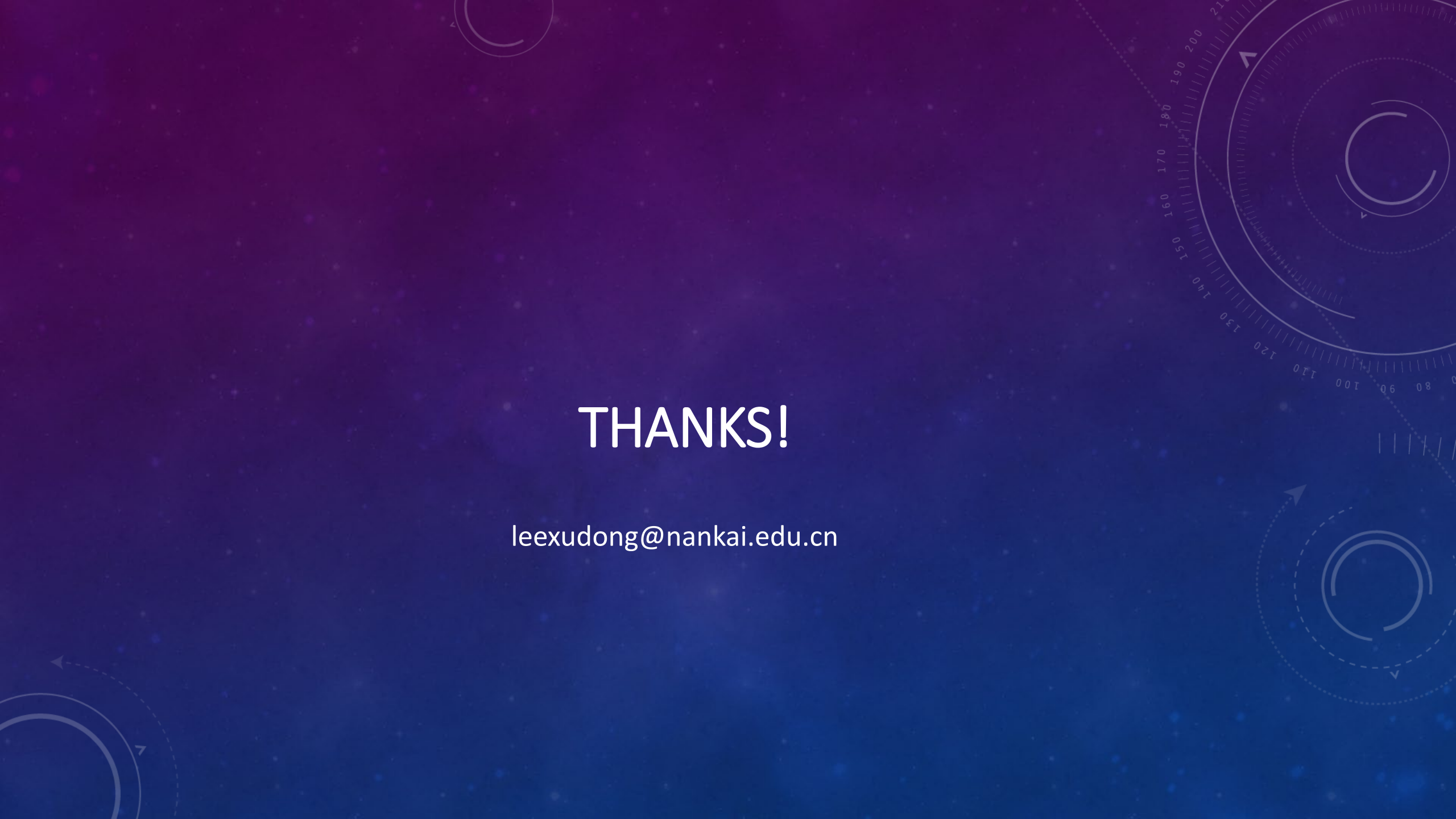
print('The result:%d' %(aCnt))
```

LAB

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THANKS!

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